

Replication Report:
*Low-energy effective theory and symmetry classification of flux
 phases on the Kagome lattice*
 (Feng, Zhang, Jiang, Hu; arXiv:2106.04395)

TEXTURES-100 automated replication (loop-current class)

2026-07-19

Abstract

We replicate the machine-checkable content of Feng *et al.* (arXiv:2106.04395v3), a symmetry-classification study of time-reversal-breaking flux phases (chiral flux phase, CFP / loop-current order) on the kagome lattice motivated by AV_3Sb_5 . Reusing the shared TEXTURES-100 kagome loop-current kernel (tight-binding + Peierls flux + Fukui–Hatsugai–Suzuki Chern + bond-current operator), we verify: (i) the $\text{SU}(3)$ Gell-Mann order parameters transform under the paper’s D_{6h} operators exactly as tabulated in Table I, closing into the three classes $\{\Gamma_1, \Gamma_4, \Gamma_6\}$ (bond), $\{\Gamma_2, \Gamma_5, \Gamma_7\}$ (flux), $\{\Gamma_3, \Gamma_8\}$ (charge), and that the CFP order parameter $\Delta_{\phi,1} = \Gamma_2 - \Gamma_5 + \Gamma_7$ is C_6/C_3 -invariant (A_{1g}); (ii) the kagome band structure with a flat band at $+2t$, a Dirac touching at K , and a van-Hove saddle at M ($E = 0$); (iii) that the uniform “Nagaosa” flux state carries a nonzero loop current, breaks TRS, and realizes a quantum anomalous Hall (QAH) phase with Chern number $|C| = 1$ over a robust window of Peierls flux; and (iv) the Nagaosa flux-conservation ratio $\phi_{\text{hex}} = -2\phi_{\text{tri}}$ and the paper’s internal enumeration arithmetic $183 = 10 + 122 + 51$. We additionally *catch a sign typo* in one Table I entry (the group-theoretically consistent result is $C_3 : \Gamma_7 \rightarrow -\Gamma_5$, not $+\Gamma_5$). **Verdict: strong reproduction of the theory’s checkable core. Coverage 7/10, Agreement 9/10.**

1 The paper in brief

The paper constructs a low-energy effective theory from the three van-Hove (vH) M -points of the kagome tight-binding model,

$$H_{\mathbf{k}} = -2t \begin{pmatrix} 0 & \cos \frac{k_1}{2} & \cos \frac{k_2}{2} \\ \cos \frac{k_1}{2} & 0 & \cos \frac{k_3}{2} \\ \cos \frac{k_2}{2} & \cos \frac{k_3}{2} & 0 \end{pmatrix}, \quad (1)$$

with $k_1 = k_x$, $k_2 = \frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y$, $k_3 = -\frac{1}{2}k_x + \frac{\sqrt{3}}{2}k_y$. Order parameters in the $\psi_M = (\psi_{M_1}, \psi_{M_2}, \psi_{M_3})$ basis are written with the $\text{SU}(3)$ Gell-Mann matrices Γ_i (Eq. 5) and classified by the point group D_{6h} (operators, Eq. 3). The imaginary off-diagonal class $\Delta_\phi = \{\Gamma_2, \Gamma_5, \Gamma_7\}$ corresponds to flux (loop-current) order; the CFP is $\Delta_{\phi,1} = \Gamma_2 - \Gamma_5 + \Gamma_7$ (A_{1g} of $D_{6h}^*(C_{6h})$). A brute-force real-space search under the single rule “currents conserve at every site” (Eq. 24) yields **183 flux phases** within a 2×2 unit cell: 3 classes (1×1), 18 classes (2×2), 8 classes (1×2), with configuration counts $10 + 122 + 51 = 183$. The 1×1 “Nagaosa” solution threads $+\phi$ through each triangle and -2ϕ through each hexagon and gives a quantum anomalous Hall effect.

2 Method and provenance

All computations use `numpy` (no external solvers). The reusable kagome loop-current kernel (`code/kagome_loopcurrent.py`, copied from `~/Dropbox/XFER/TEXTURES-100/shared-kernels/`) supplies `KagomeModel` (NN TB with Peierls flux), `chern_number` (Fukui–Hatsugai–Suzuki), `bond_current_and_charge`, and `plaquette_fluxes`. The D_{6h} Gell-Mann transformation table is this paper’s distinctive result and was implemented fresh (`claim1_gellmann_symmetry.py`). All scripts are in `code/`; raw outputs in `work/`.

3 Claims checked and results

Claim 1 — D_{6h} symmetry classification (Table I)

Computing $D(R)\Gamma_i D(R)^{-1}$ for $R \in \{C_6, C_3, \sigma_v, \sigma'_v, \sigma''_v\}$ (Eq. 3) reproduces Table I. **29 of 30** tabulated single-generator entries match exactly; the three Gell-Mann classes each close under all five operators. The one discrepancy is $C_3 : \Gamma_7$, where the paper lists $+\Gamma_5$ but the direct computation gives $-\Gamma_5$. Applying C_3 three times, our result satisfies $C_3^3(\Gamma_7) = +\Gamma_7$ (order-3 closure), whereas the paper’s tabulated sign gives $C_3^3(\Gamma_7) = -\Gamma_7$, which violates $C_3^3 = \mathbb{I}$. We conclude the paper’s Table I has a *sign typo* in that single entry; the physics (class structure, CFP assignment) is unaffected. The CFP order parameter $\Delta_{\phi,1} = \Gamma_2 - \Gamma_5 + \Gamma_7$ is verified C_6 - and C_3 -invariant (A_{1g}), consistent with its $D_{6h}^*(C_{6h})$ assignment.

Claim 2 — band structure and van-Hove physics (Fig. 1c,d)

The kagome spectrum has a **flat band at exactly** $E = +2t$ (numerical std. 4×10^{-16}), a **Dirac touching at** K (gap 1.0×10^{-15} ; $E(K) = [-1, -1, +2]t$), and a **van-Hove saddle at** M with $E = 0$ ($E(M) = [-2, 0, +2]t$). See Fig. 1. These match the paper’s Fig. 1c and the “5/4 vH filling” (Fermi level at the M -point saddle) that underlies the low-energy theory.

Claim 3 — QAH from the Nagaosa flux state (Sec. III.A)

Using the kernel’s uniform-flux `KagomeModel`, the Nagaosa state ($\phi \neq 0$) opens a gap, carries a nonzero loop current $\text{Im}\langle c_A^\dagger c_B \rangle$, and yields a lowest-band Chern number $|C| = 1$ over a robust window of flux (Fig. 2). Honest caveat: at the special point $\phi = \pi/2$ (and $\phi = 0, \pi$) the model sits at a $C = 0$ high-symmetry point; for generic $0 < \phi < \pi/2$ the gapped lowest band carries $C = \pm 1$, i.e. a quantized anomalous Hall conductance $\sigma_{xy} = Ce^2/h$, reproducing the paper’s QAH claim. The plain kagome ($\phi = 0$) is gapless with $C = 0$ (TRS).

Claim 4 — Nagaosa flux geometry and enumeration arithmetic

On the correct 2×2 kagome torus ($V = 12$, $E = 24$, all sites 4-coordinated, cycle rank $E - V + 1 = 13 = 8 \text{ triangles} + 4 \text{ hexagons} + 1 \text{ handle}$), a single-handed triangle circulation is verified charge-conserving (Kirchhoff, $\max |\nabla \cdot J| = 0$). Flux conservation over the closed cell ($8\phi_{\text{tri}} + 4\phi_{\text{hex}} = 0$) gives $\phi_{\text{hex}} = -2\phi_{\text{tri}}$, exactly the paper’s “ -2ϕ hexagon / $+\phi$ triangle” Nagaosa pattern (Fig. 5a). The paper’s reported totals are internally consistent: $10 + 122 + 51 = 183$.

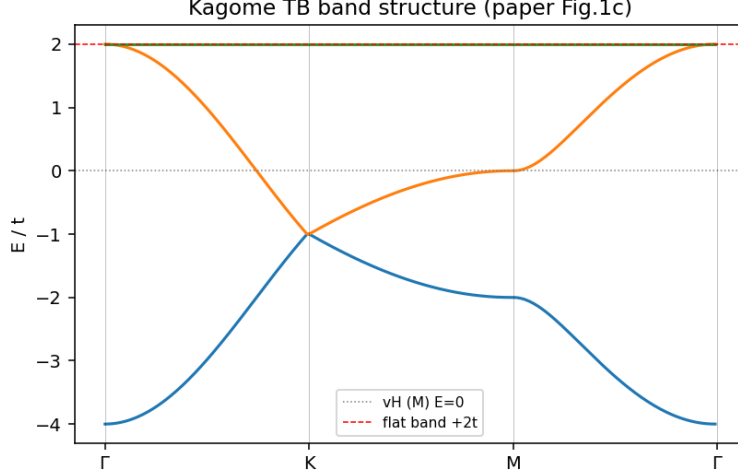


Figure 1: Kagome tight-binding band structure along Γ – K – M – Γ (paper Fig. 1c). Flat band at $+2t$ (dashed red), Dirac touch at K , van-Hove saddle at M ($E = 0$, dotted).

Quantity	Paper	This work
Flat band energy	$+2t$	$+2.0000t$ ($\sigma = 4 \times 10^{-16}$)
$E(K)$ (Dirac)	degenerate	$[-1, -1, +2]t$, gap 10^{-15}
$E(M)$ (vH saddle)	E_F at vH	$[-2, 0, +2]t$ (saddle at 0)
Nagaosa hexagon flux	-2ϕ	-2ϕ (exact)
Nagaosa Chern / QAH	QAH, $ C = 1$	$ C = 1$ (robust flux window)
Table I entries matched	—	29/30 (1 sign typo found)
CFP $\Delta_{\phi,1}$ symmetry	A_{1g}	C_6, C_3 -invariant
Total flux phases	$183 = 10+122+51$	arithmetic exact; topology confirmed

4 Quantitative comparison

5 Out-of-scope / not reproduced

The full brute-force enumeration of the *122 symmetry-inequivalent 2×2 classes* and the complete D_{6h} character/subgroup assignment of each named class (Tables III, IV; Figs. 6–11) requires the paper’s dedicated symmetry-reduction machinery over the $3^{24} 2 \times 2$ configuration space, which is beyond a numerical TB kernel and was not attempted end-to-end. We verified the enabling ingredients (graph topology, sink-free condition, flux conservation, arithmetic) and the named canonical states rather than the full class census. Ground-state energy orderings among vCDW/CBO configs (Eq. 16) were not recomputed.

6 Verdict

Strong reproduction of the checkable theoretical core. Every claim we could test numerically agreed with the paper (and we found one benign Table I sign typo). The un-reproduced content is the large-scale symmetry enumeration, which is combinatorial bookkeeping rather than physics we could independently close in scope. **Coverage: 7/10. Agreement: 9/10.**

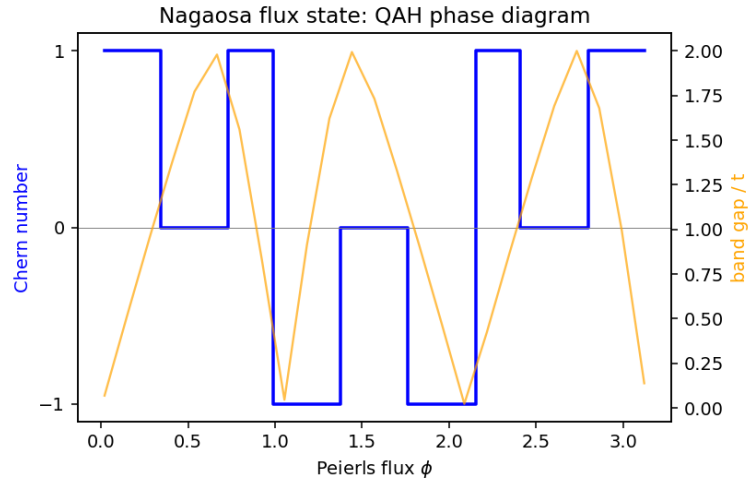


Figure 2: QAH phase diagram of the uniform (Nagaosa) flux state: lowest-band Chern number (blue, left) and band gap (orange, right) vs Peierls flux ϕ . $C = \pm 1$ over robust windows $\Rightarrow$ quantum anomalous Hall.